

Supplementary Material for “RTNet: A Multi-scale Multi-receptive-field Multi-attention Fusion Approach for Malignancy Risk Assessment of Pulmonary Nodules”

This supplementary material provides rigorous mathematical derivations and exhaustive experimental breakdowns that complement the main manuscript. While the main text focuses on the clinical applicability and overall architectural innovations, this document details the low-level tensor operations and per-class performance variances for reproducibility purposes.

I. S-I: DETAILED DATA PREPROCESSING PIPELINE

Data preprocessing follows three standardized stages (illustrated in Fig. 1 of the main manuscript):

Stage 1: XML annotation processing - DICOM SOPInstanceUID fields are matched with XML imageSOP_UID fields for precise image-annotation alignment. CT Hounsfield Unit (HU) values $[-1000, 2000]$ are linearly normalized to $[0, 255]$ and converted to RGB JPEG format for conventional convolution operations.

The HU normalization is mathematically defined as:

$$X = \frac{X_{\text{HU}} + 1000}{3000} \times 255 \quad (1)$$

where X denotes the normalized HU values and X_{HU} represents the original HU intensities.

Stage 2: Multi-reader consensus - Malignancy scores from four radiologists are extracted via XML readingSession nodes. Final labels are determined by majority voting, discarding samples where majority proportion $< 66.7\%$ to ensure annotation reliability. For a panel of four radiologists, a majority proportion of 66.7% corresponds to agreement from at least three radiologists ($3/4 = 75\% \geq 66.7\%$). Thus, samples were only retained if at least three radiologists agreed on the malignancy score. Samples with only two agreeing radiologists (50% agreement) or less were discarded, as this does not meet the 66.7% threshold.

Stage 3: Spatial normalization - Nodule contour coordinates from XML edgeMap nodes are processed to compute minimum axis-aligned bounding boxes (AABB). Box center coordinates and dimensions are normalized to $[0, 1]$ range. To address the spatial continuity of nodules exceeding 3mm in diameter, we extract three consecutive axial slices centered at the radiologist-annotated nodule centroid. These slices are stacked along the channel dimension to form a multi-slice representation (size: $224 \times 224 \times 3$), where each channel

corresponds to one axial slice. This design preserves 3D contextual information while maintaining compatibility with 2D convolutional architectures. Each TXT file entry corresponds to a nodule instance (class label + normalized bounding box). OpenCV [1] visual verification ensures spatial congruence between boxes and nodule sizes, with manual correction for anomalies.

II. S-II: DETAILED COMPUTATIONAL COMPLEXITY ANALYSIS

This section provides the exhaustive computational cost breakdown that is summarized in the main text. All FLOPs and parameter counts were measured under the same hardware and input resolution ($224 \times 224 \times 3$) to ensure fair comparison.

To investigate the computational complexity of our model, for the RTNet model and TMFM module, we used the torchinfo tool in the PyTorch framework to calculate the floating-point operations and parameter count. As shown in Table S-III, with an input dimension of $224 \times 224 \times 3$, the FLOPs and the Params count of the RTNet model are 4.94 GFLOPs and 78.98 million, respectively, compared to 3.68 GFLOPs and 21.29 million for its backbone network ResNet34. RTNet introduces an additional computational cost of 34.2% in FLOPs ($3.68 \rightarrow 4.94$ GFLOPs) and 270.9% in parameter count ($21.29 \rightarrow 78.98$ million) compared to the ResNet34 backbone. This increased capacity enables three key architectural innovations within the TMFM module: 1) Multi-scale dilated convolutions expand receptive fields to cover all nodule sizes in LIDC-IDRI; 2) Median pooling enhances feature statistic robustness; 3) Pixel-attention guided dynamic fusion improves multi-scale feature consistency. Experimental results show RTNet achieves 86.14% weighted accuracy (34.7% improvement over ResNet-34) with real-time inference, validating its technical superiority in fine-grained risk assessment.

Taking the TMFM module in the fourth layer (RT Layer4) of RTNet as an example (input tensor dimensions: $7 \times 7 \times 512$), Table S-IV summarizes the floating-point operations (FLOPs), parameter count, and memory footprint of each component within the TMFM module. In this module, the Multi-Scale Multi-Receptive Field block accounts for 42.3% of the total FLOPs and 39.9% of the parameters. This is followed by

TABLE S-I
COMPUTATIONAL COMPLEXITY COMPARISON BETWEEN RTNET AND RESNET-34.

Methods	FLOPs (GFLOPs)	Params (M)
ResNet34	3.68	21.29
RTNet (Ours)	4.94	78.98

the Median Channel Attention module (16.6% FLOPs and 20.0% parameters), Median Spatial Attention module (21.6% FLOPs and 23.9% parameters), and Pixel Attention-Guided Feature Fusion module (19.5% FLOPs and 16.2% parameters). Notably, the Pixel Attention-Guided Feature Fusion module efficiently integrates three attention feature tensors while consuming only 183.98 MFLOPs (19.5% of total module FLOPs) and 3.88 MB of parameters (16.2% of total module parameters), thereby providing an efficient and robust feature fusion solution for the TMFM module.

TABLE S-II
DETAILED FLOPs AND PARAMETERS BREAKDOWN OF SUB-MODULES WITHIN TMFM (RT LAYER4). MMCM: MULTI-SCALE MULTI-RECEPTIVE-FIELD CONVOLUTION MODULE; MCAM: MEDIAN-ENHANCED CHANNEL ATTENTION MECHANISMS; MSAM: MEDIAN-ENHANCED SPATIAL ATTENTION MECHANISMS; PAFFM: PIXEL-ATTENTION GUIDED FEATURE FUSION MECHANISMS.

Modules	FLOPs (MFLOPs)	Params (M)
MMCM	398.72	9.55
MCAM	156.18	4.78
MSAM	203.57	5.73
PAFFM	183.98	3.88

III. S-III: DETAILED MATHEMATICAL FORMULATIONS OF TMFM SUB-MODULES

This section provides the complete tensor-level derivations for the median-enhanced attention mechanisms and pixel-attention guided fusion module, as referenced in the main text (Section II-C).

A. Median-Enhanced Channel and Spatial Attention Mechanisms

Different channels in convolutional neural networks (CNNs) often exhibit distinct feature As illustrated in Fig. 3c of the main text, the median-enhanced channel attention mechanism employs three parallel pooling operations along the channel dimension: average pooling F_{avg} , max pooling F_{max} , and the proposed median pooling F_{median} . For an input feature map $\mathbf{X} \in \mathbb{R}^{C \times H \times W}$, these operations generate compressed channel descriptors:

$$\begin{aligned} F_{avg} &= \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W \mathbf{X}[:, i, j], \\ F_{max} &= \max_{i,j} \mathbf{X}[:, i, j], \\ F_{median} &= \text{median}_{i,j} \mathbf{X}[:, i, j] \end{aligned} \quad (\text{S1})$$

Each descriptor is then processed through an expand function containing two fully connected (FC) layers with ReLU activation, producing refined channel weights $\mathbf{Y}_1, \mathbf{Y}_2$, and $\mathbf{Y}_3$. The

final channel attention map $\mathbf{Y}_C$ is obtained through element-wise summation:

$$\mathbf{Y}_C = \sigma(\mathbf{Y}_1 + \mathbf{Y}_2 + \mathbf{Y}_3) \otimes \mathbf{X} \quad (\text{S2})$$

where σ denotes the sigmoid function and $\otimes$ represents channel-wise multiplication.

The spatial attention counterpart (Fig. 3d of the main text) operates through complementary pooling along spatial axes. For each channel c we compute:

$$\begin{aligned} X_{avg} &= \frac{1}{C} \sum_{k=1}^C \mathbf{X}[k, \cdot], \\ X_{max} &= \max_k \mathbf{X}[k, \cdot], \\ X_{median} &= \text{median}_k \mathbf{X}[k, \cdot] \end{aligned} \quad (\text{S3})$$

These spatial descriptors are concatenated and processed by a 1×1 convolution layer followed by sigmoid normalization, generating the spatial weight map $\mathbf{Y}_S \in \mathbb{R}^{1 \times H \times W}$:

$$\mathbf{Y}_S = \sigma(\text{Conv}_{1 \times 1}([X_{avg}; X_{max}; X_{median}])) \otimes \mathbf{X} \quad (\text{S4})$$

This dual attention design achieves synergistic noise suppression—channel attention eliminates feature channel outliers while spatial attention filters irrelevant background regions. The median pooling component enhances robustness to CT imaging artifacts compared to conventional max/average pooling approaches.

B. Pixel-Attention Guided Feature Fusion

As depicted in Fig. 3e of the main text, the pixel attention fusion module operates through two principal phases: pixel-wise global context extraction and hierarchical feature integration. Given input features $\mathbf{X} \in \mathbb{R}^{C \times H \times W}$ from preceding layers and intermediate features $\mathbf{P1} \in \mathbb{R}^{C' \times H \times W}$, we first perform dimensional transformation:

$$\mathbf{S} = \text{Reshape}(\text{Concat}(\text{Permute}(\mathbf{X}), \text{Permute}(\mathbf{P1}))) \in \mathbb{R}^{(C+C') \times (H \times W)} \quad (\text{S5})$$

where $\text{Permute}()$ switches tensor dimensions to $[H, W, C]$ and $\text{Reshape}()$ flattens spatial dimensions. The concatenated tensor $\mathbf{S}$ undergoes pixel attention generation:

$$\mathbf{A}_p = \sigma(\text{Conv}_{1 \times 1}(\mathbf{S})) \in \mathbb{R}^{1 \times H \times W} \quad (\text{S6})$$

with σ denoting the sigmoid activation. This attention map modulates feature importance at individual pixel locations through element-wise multiplication:

$$\mathbf{Y}_P = \mathbf{A}_p \otimes \mathbf{X} \quad (\text{S7})$$

The final fusion output (Fig. 3b of the main text) combines multi-scale features through channel compression:

$$\mathbf{Y} = \text{Conv}_{1 \times 1}(\mathbf{X} + \mathbf{Y}_P \otimes (\mathbf{Y}_C + \mathbf{Y}_S)) \quad (\text{S8})$$

This architecture achieves three critical advantages: 1) The 1×1 convolution in (S8) reduces channel dimensions from $C + C'$ to C , halving computational costs while maintaining representational capacity; 2) Global pixel attention weighting enhances focus on malignancy-related regions; 3) Cross-scale feature aggregation preserves both high-frequency textures from shallower layers and semantic abstractions from deeper layers.

IV. S-IV: FULL IMPLEMENTATION DETAILS AND METRIC DEFINITIONS

A. Implementation Details

The experiments were conducted on hardware equipped with NVIDIA RTX3080 GPU (10GB VRAM), accelerated by PyTorch 2.0.1 [2] and CUDA 11.7. The dataset was randomly partitioned into test/train sets at a 1:9 ratio, with the final epoch model evaluated on the test set for performance metrics. The test set comprised 415 cases with risk levels distribution: Grade 1 (n=57), Grade 2 (n=116), Grade 3 (n=144), Grade 4 (n=52), and Grade 5 (n=46). To address class imbalance, focal loss was employed with inverse class frequency weighting and a focusing parameter $\gamma=1.5$ to mitigate majority-class bias [3]. Input images were resized to 224×224 pixels. AdamW optimizer [4] was used with initial learning rate 5×10^{-5} and cosine annealing scheduler (T_max=64 cycles) [5] over 500 epochs to dynamically adjust learning rates for convergence-stability trade-off. Training configuration included batch size 16, gradient clipping (threshold=1.0), and label smoothing ($\alpha=0.1$) to prevent overfitting. For reproducibility, random seeds (1-10000 range) were fixed with mixed-precision training enabled. The complete experimentation required approximately 104 hours.

B. Evaluation Metrics

1) *Weighted Accuracy (W-Acc)*: *Weighted Accuracy (W-Acc)* is a classification accuracy metric that accounts for class distribution disparities. It computes the overall accuracy through class-specific sample weighting, formally defined as:

$$W\text{-Acc} = \sum_{i=1}^N \frac{n_i}{n_{\text{total}}} \times \text{Acc}_i \quad (2)$$

Here, $N = 5$ denotes the total number of risk levels, n_i represents the sample count of grade i , n_{total} is the total sample size, and Acc_i is the classification accuracy for grade i . This metric mitigates the bias of conventional accuracy in class-imbalanced datasets by enforcing balanced performance across all risk strata. For instance, if a model completely misclassifies minority classes, its W-Acc will dramatically decrease, whereas conventional accuracy might remain inflated due to majority-class overfitting. In our study, the natural imbalance across risk levels necessitates W-Acc to objectively assess model fidelity.

2) *Average Precision (AP)*: *Average Precision (AP)*, defined as the area under the Precision-Recall (PR) curve, quantifies model robustness across decision thresholds:

$$AP = \int_0^1 P(R) dR \quad (3)$$

where R denotes recall and $P(R)$ is the precision at recall level R . AP is particularly suited for multi-class scenarios due to its sensitivity to minority classes and capacity to characterize inter-class ambiguity. Compared to single-threshold precision metrics, AP comprehensively evaluates model performance across varying decision boundaries, making it ideal for imbalanced medical datasets. In our study, AP primarily measures

RTNet's capability to discriminate positive vs. negative samples across risk levels. Higher AP values indicate more stable classification outcomes and improved capture of fine-grained malignancy signatures.

3) *Area Under the ROC Curve (AUC)*: *Area Under the ROC Curve (AUC)* evaluates the model's global discriminative capability by measuring the area under the Receiver Operating Characteristic (ROC) curve. With False Positive Rate (FPR) as the x-axis and True Positive Rate (TPR, equivalent to recall) as the y-axis, AUC is formally defined as:

$$AUC = \int_0^1 \text{TPR}(\text{FPR}) d\text{FPR} \quad (4)$$

For multi-class scenarios, we employ the "one-vs-one" AUC extension to assess pairwise class discriminability. Ranging from 0 to 1, higher AUC values indicate superior classification performance. In our study, AUC explicitly quantifies RTNet's ability to differentiate risk levels, particularly in ambiguous cases:

V. S-V: EXTENDED EXPERIMENTAL RESULTS AND ABLATION DETAIL

1) *Comparative Analysis*: To further validate the superiority of RTNet in fine-grained malignancy risk assessment of pulmonary nodules, we conducted comparative experiments using different reproducible backbone network architectures. Seven mainstream deep learning models were selected: ResNet34, ResNet50, ResNet101, ResNet152, DenseNet121, DenseNet264, and Transformer-based ViT. Systematic comparisons across three metrics (W-Acc, AUC, AP) demonstrate the effectiveness of our approach. Tables S-III-S-V present the performance of all models on these metrics. The results show that RTNet comprehensively outperforms existing methods in the five-class classification task, with particularly significant advantages in noise-sensitive scenarios such as intermediate risk levels (Grades 2-3).

As shown in Table S-III, RTNet achieves a total weighted accuracy (W-Acc) of 86.14% in the five-class classification task, comprehensively outperforming all baseline models. It surpasses the second-best model DenseNet264 DenseNet264 by 16.26 percentage points (a relative improvement of 23.27%) and significantly exceeds ViT. For the fundamental challenges of malignancy classification — Grade 1 (benign) and Grade 5 (highly malignant), where the pathological distinctions are most prominent — RTNet attains the highest accuracy among all eight models.

In addressing the critical challenge of intermediate-risk classification (Grades 2-4), RTNet demonstrates notable superiority: In grade 2: Accuracy of **82.93%** (an absolute increase of 35.99 percentage points over ResNet34 and 18.64 percentage points over DenseNet264). In grade 3: Accuracy of **89.06%** (an absolute increase of 22.92 percentage points over ResNet34 and 14.68 percentage points over DenseNet264). In grade 4: Accuracy of **81.48%**, trailing the optimal model ResNet101 (84.62%) by only 3.14 percentage points.

TABLE S-III
CLASSIFICATION ACCURACY COMPARISON (W-ACC). BEST RESULTS IN BOLD.

Methods	Grade 1	Grade 2	Grade 3	Grade 4	Grade 5	W-Acc (%)
ResNet34	0.8125	0.4694	0.6614	0.7593	0.7179	62.89
ResNet50	0.8103	0.4493	0.6694	0.6727	0.9070	64.11
ResNet101	0.7534	0.4711	0.5984	0.8462	0.9524	65.54
ResNet152	0.7500	0.5676	0.6934	0.7544	0.7857	68.67
DenseNet121	0.6053	0.4747	0.6522	0.7818	0.7347	67.27
DenseNet264	0.8095	0.6429	0.7438	0.5122	0.9459	69.88
ViT	0.2276	0.3281	0.5000	0.1579	0.3953	29.88
RTNet(Ours)	0.8209	0.8293	0.8906	0.8148	0.9535	86.14
Support	57	116	144	52	46	415

TABLE S-IV
AUC COMPARISON. BEST RESULTS IN BOLD. N/A INDICATES THAT DATA IS NOT AVAILABLE.

Methods	Grade 1	Grade 2	Grade 3	Grade 4	Grade 5	W-AUC
ResNet34	0.9353	0.8317	0.7973	0.9239	0.9833	0.8446
ResNet50	0.9644	0.8317	0.8019	0.9183	0.9758	0.8519
ResNet101	0.9786	0.8362	0.7919	0.9345	0.9947	0.8583
ResNet152	0.9592	0.8618	0.8091	0.9246	0.9739	0.8657
DenseNet121	0.9302	0.8216	0.7759	0.9200	0.9688	0.8371
DenseNet264	0.9347	0.9036	0.8755	0.9116	0.9296	0.8985
ViT	0.5808	0.6652	0.5655	0.6122	0.8063	0.6427
CNN Model [6]	N/A	N/A	N/A	N/A	N/A	0.78
RNN Model [6]	N/A	N/A	N/A	N/A	N/A	0.78
Cancer risk evaluation module [6]	N/A	N/A	N/A	N/A	N/A	0.86
RTNet(Ours)	0.9968	0.9652	0.9622	0.9576	0.9852	0.9635
Support	57	116	144	52	46	N/A

TABLE S-V
AP COMPARISON. BEST RESULTS IN BOLD.

Methods	Grade 1	Grade 2	Grade 3	Grade 4	Grade 5
ResNet34	0.8192	0.5389	0.6972	0.7934	0.7728
ResNet50	0.8669	0.5781	0.6949	0.7127	0.9327
ResNet101	0.9094	0.5391	0.6411	0.8397	0.9684
ResNet152	0.8741	0.6529	0.7132	0.8034	0.9112
DenseNet121	0.7514	0.9582	0.6819	0.7745	0.8342
DenseNet264	0.8629	0.7988	0.7669	0.6857	0.8646
ViT	0.2292	0.3634	0.4198	0.2305	0.4303
RTNet(Ours)	0.9846	0.9348	0.9078	0.9025	0.9457
Support	57	116	144	52	46

These results directly validate the capability of the Triple Multi-Fusion Module (TMFM) to capture ambiguous pathological features, enabling RTNet to achieve substantial accuracy improvements in intermediate-risk levels (2-3) compared to other baselines. Furthermore, RTNet maintains competitive accuracy across all risk levels, affirming its robustness in comprehensive pulmonary nodule risk assessment.

As evidenced by Table S-IV, RTNet achieves the highest AUC values across Grades 1–4, demonstrating exceptional performance in distinguishing low-to-intermediate malignancy risks. Notably, in Grade 3—a category with high diagnostic ambiguity—RTNet maintains an AUC of **0.9622**, whereas all baseline models fail to surpass **0.9**, confirming its clinical reliability for malignancy risk stratification. For Grade 5 nodules, RTNet’s AUC of **0.9852** slightly trails ResNet101’s peak value (**0.9947**); however, when combined with AP metrics from Table S-V, RTNet exhibits superior stability under noisy conditions. In low-risk levels, RTNet demonstrates outstanding discriminative performance. For Grade 1, RTNet achieves an AUC of **0.9968**, outperforming ResNet101 (**0.9786**) by

0.0182 and DenseNet264 (**0.9347**) by 0.0621. For Grade 2, RTNet achieves an AUC of **0.9652**, surpassing ResNet101 (**0.8362**) by 0.1290 and DenseNet264 (**0.9036**) by 0.0616. These results consistently confirm RTNet’s superior capability in distinguishing benign and low-risk nodules from higher-risk categories. Despite ResNet101’s near-parity with RTNet in Grade 1 AUC, its significantly lower AP reflects deficiencies in modeling benign nodule characteristics due to limited feature scales.

Table S-V further corroborates RTNet’s robustness, with AP values exceeding ResNet34 across all grades. While DenseNet121 achieves a marginally higher AP (**0.9582**) in Grade 2 compared to RTNet’s **0.9348**, its discordantly low accuracy (as per Table S-IV) suggests overfitting and poor generalization. In contrast, RTNet’s multi-attention synergy ensures congruent improvements in both AP and accuracy. For Grades 3 and 4, RTNet’s AP substantially outperforms all competitors, and in Grade 5, its AP of **0.9457** closely approaches ResNet101’s optimal **0.9684**.

To evaluate the comprehensive performance of RTNet, comparative analysis was conducted against: CNN Model, RNN Model and Cancer Risk Evaluation Module proposed by Wang et al. [6] (detailed in Table S-V). Empirical results demonstrate RTNet’s statistically superior weighted average AUC (0.9635) over the benchmark models in Wang’s [6], confirming its advanced discriminative capability. Critically, RTNet is the sole approach providing fine-grained AUC stratification. It achieves outstanding performance across all five risk levels, whereas Wang et al. [6] only report global AUC, precluding grade-specific discriminative assessment. This granular advantage holds critical clinical value: RTNet’s consistently ultra-high AUC per grade establishes a reliable foundation for

TABLE S-VI
CLASSIFICATION ACCURACY OF ABLATION MODELS (W-ACC). BEST RESULTS IN BOLD.

Methods	Grade 1	Grade 2	Grade 3	Grade 4	Grade 5	W-Acc (%)
RTNet_C	0.8276	0.8333	0.6614	0.7500	0.3925	74.68
RTNet_P	0.7183	0.7423	0.8722	0.7606	0.7907	79.47
RTNet_S	0.7778	0.7460	0.8083	0.7742	0.8636	78.85
RTNet_T	0.7288	0.7857	0.8413	0.8082	0.9111	81.20
RTNet_H	0.7869	0.4713	0.6577	0.5588	0.8519	65.16
RTNet_C_M	0.7391	0.8333	0.8261	0.4286	0.7708	76.25
RTNet_S_M	0.8358	0.7043	0.8551	0.7647	0.8182	80.07
RTNet(Ours)	0.8209	0.8293	0.8906	0.8148	0.9535	86.14
Support	57	116	144	52	46	415

TABLE S-VII
AUC OF ABLATION MODELS. BEST IN BOLD.

Methods	Grade 1	Grade 2	Grade 3	Grade 4	Grade 5
RTNet_C	0.9855	0.9104	0.8653	0.9232	0.9785
RTNet_P	0.9661	0.9315	0.9397	0.9382	0.9662
RTNet_S	0.9727	0.9504	0.9141	0.8949	0.9909
RTNet_T	0.9714	0.9559	0.9287	0.9867	0.9946
RTNet_H	0.9420	0.8195	0.8066	0.9175	0.9864
RTNet_C_M	0.9846	0.8969	0.8847	0.9015	0.9838
RTNet_S_M	0.9897	0.9311	0.9214	0.9554	0.9787
RTNet(Ours)	0.9968	0.9652	0.9622	0.9576	0.9852
Support	57	116	144	52	46

TABLE S-VIII
AP OF ABLATION MODELS. BEST IN BOLD.

Methods	Grade 1	Grade 2	Grade 3	Grade 4	Grade 5
RTNet_C	0.9116	0.8018	0.8035	0.7258	0.9071
RTNet_P	0.8918	0.8442	0.8993	0.8522	0.8378
RTNet_S	0.9064	0.8827	0.8456	0.8177	0.9415
RTNet_T	0.8835	0.9024	0.8888	0.9423	0.9666
RTNet_H	0.8247	0.4814	0.6694	0.6625	0.9484
RTNet_C_M	0.8904	0.7813	0.8355	0.7262	0.9060
RTNet_S_M	0.9538	0.8423	0.8880	0.8761	0.8823
RTNet(Ours)	0.9846	0.9348	0.9078	0.9025	0.9457
Support	57	116	144	52	46

individualized prognostic assessment.

2) *Ablation Study Results*: To validate the contributions of the TMFM module and its internal mechanisms to RTNet’s performance, we designed five ablation experiments quantifying impacts through three dimensions: attention mechanism integrity, feature fusion strategy, and module placement - involving both positional adjustments and internal component removal. Specifically, we constructed five variant models: (1) RTNet_C: Removed channel attention while retaining spatial attention. (2) RTNet_P: Replaced the fusion module with direct summation of spatial/channel features followed by point-wise multiplication with original features. (3) RTNet_S: Removed spatial attention while retaining channel attention. (4) RTNet_T: Embedded TMFM only after the final residual layer. (5) RTNet_H: Embedded TMFM only after the first residual layer. (7) RTNet_S_M: Removes median pooling in the spatial attention branch while retaining average and max pooling operations. (8) RTNet_C_M: Removes median pooling in the channel attention branch while retaining average and max pooling operations.

As shown in Table S-VI, RTNet_C, RTNet_S, and RTNet_P suffered significant performance degradation, with weighted accuracy dropping to **74.68%**, **78.85%**, and **79.47%**, respec-

tively. Notably, RTNet_C exhibited catastrophic failure in classifying high-grade malignant nodules (Grade 5), with accuracy dropping from **95.35%** (full RTNet) to just **39.25%** — a dramatic decline of **56.1** percentage points. This stark degradation highlights the critical role of median-enhanced channel attention in high-risk levels discrimination. Intriguingly, RTNet_C marginally improved accuracy in low-risk levels 1–2, though this may result from stochastic training variance. Simultaneously, Table S-VII reveals that RTNet_C’s AUC for Grade 3 nodules dropped from **0.9622** to **0.8653**, a decrease of 0.0969, indicating impaired classification of ambiguous cases. This decline corroborates the channel attention’s core function in noise suppression and inter-class discriminability enhancement. Furthermore, Table S-VIII demonstrates that RTNet_C’s AP for Grade 5 nodules decreased from **0.9457** to **0.9071** (a reduction of 0.0386), reflecting destabilized precision-recall balance in distinguishing aggressive malignancies.

RTNet_S, devoid of spatial attention, exhibited accuracy reductions across all grades (Table S-VI). The spatial weight maps generated by the median-enhanced spatial attention encapsulate critical nodule characteristics across risk levels. Its removal caused an AP drop of 0.0622 for Grade 3 nodules, attributable to insufficient localization of subtle spatial patterns (e.g., micronodular spiculations). Paradoxically, while RTNet_S achieved an abnormally high AUC of 0.9909 for Grade 5 nodules, its overall weighted accuracy decreased by **7.3%** due to over-reliance on channel statistics and loss of contextual spatial cues, resulting in false positives in holistic classification.

The dynamic weight allocation capability of the feature fusion module was validated across all metrics (Tables S-VI–S-VIII). RTNet_P exhibited accuracy declines in Grades 1, 2, 4, and 5, most pronounced in Grades 1 and 5. Although Grade 3 accuracy remained comparable to the full RTNet, synchronized AP and AUC degradation exposed the limitations of naive feature addition—loss of high-frequency wavelet details and multi-scale receptive field representations. For Grade 2 nodules, RTNet_P’s AP of **0.8442** represents a decrease of 0.0906 from the full model. In contrast, the complete RTNet, with pixel-level attention recalibration, achieved synchronized improvements (AP: **0.9348**, Accuracy: **82.93%**), demonstrating the fusion module’s efficacy in feature refinement and multi-attention equilibrium.

To comprehensively validate the noise suppression capa-

bility of median pooling, we introduced two specialized ablation models, RTNet_C_M and RTNet_S_M, for training. Quantitative results in Tables S-VI–S-VIII show significant performance degradation in both models compared to the full RTNet. Specifically, RTNet_C_M shows a decrease of 6.45% in accuracy for diagnostically challenging grade 3 nodules, while RTNet_S_M shows a reduction of 3.55% in accuracy for the same category, directly attributable to weakened artifact suppression in thoracic CT imaging. For high-risk level 5 malignant nodules, the absence of median pooling in channel attention causes accuracy to plummet to 77.08%, and grade 4 malignant’s nodules accuracy to plummet to 42.86%, indicating increased sensitivity to high-intensity outliers. These empirical results quantitatively confirm that median pooling contributes critical performance improvements in noise-rich diagnostic environments, establishing its indispensable role in robust pulmonary nodule risk stratification.

The depth-dependent effects of module embedding are notably pronounced across Tables S-VI–S-VIII. RTNet_T, embedded with TMFM only in deep layers, achieved over **10%** higher accuracy than RTNet_H (shallow-layer embedded) in Grades 2–5 malignancy classification. Strikingly, its AP value (0.9423) surpassed RTNet_H by **42%**, highlighting the critical role of deep semantic features in decoding advanced pathological markers. Additionally, RTNet_H’s AP of 0.8247 in Grade 1 nodules accounted for merely **83.7%** of the full model’s performance, indicating that shallow embeddings fail to support the semantic abstraction required for fine-grained malignancy classification. Despite these gains, RTNet_T’s AUC (0.9559) in Grade 2 remained inferior to the full RTNet, emphasizing the necessity of integrating multi-scale shallow features to bridge performance gaps. The hierarchical complementarity strategy—where shallow TMFM layers capture local textures and deep layers aggregate global semantics—validates the superiority of four-layer TMFM embedding in the complete RTNet architecture.

VI. S-VI: EXTENDED DISCUSSION ON ARCHITECTURAL MOTIVATIONS

As referenced in the main text, this section provides the comprehensive architectural rationale that underlies the TMFM module design.

The TMFM module acquires malignant features through three principal pathways:

1) Integration of Multi-Scale Dilated Convolutions and Wavelet Convolutions: Hybrid multi-scale dilated and wavelet convolutions extract feature tensors with varying receptive fields from three dilation rates (3/5/7) and db1 wavelet transforms. Feature concatenation and fusion enhance sensitivity to nodule size variations and global context, overcoming the limited adaptability of fixed-receptive-field approaches.

2) Median Pooling Enhanced Feature Extraction: A median pooling enhanced feature extraction method synergizes with max/average pooling in dual attention mechanisms (channel

and spatial), improving feature comprehensiveness and noise robustness during malignancy characterization.

3) Pixel Attention Guided Feature Fusion: A pixel-attention guided fusion module dynamically weights features across channel, spatial, and pixel dimensions, preserving global context while enabling cross-dimensional feature optimization for improved classification precision.

Furthermore, we conducted comprehensive ablation studies on RTNet, categorized into two groups:

Depth-aware TMFM Embedding: By reducing the number of embedded TMFM layers, we constructed two variants: RTNet.H (retains only the first residual layer with TMFM modules) and RTNet.T (retains only the last residual layer with TMFM modules).

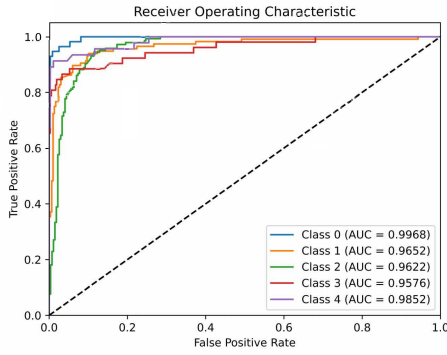
Component Ablation Analysis: To optimize training efficiency, we generated three variants based on RTNet.T: RTNet.C showed acceptable performance in low-risk levels 1–2 but experienced significant declines in high-risk levels 3–5, with accuracy and AUC plunging by over 25% in Grades 4–5. This validates the noise suppression capability of median-enhanced channel attention for high-grade malignancies. RTNet.S exhibited reduced accuracy across all categories, proving that median-enhanced spatial attention universally enhances classification by capturing nodule boundary and morphological patterns. RTNet.P underperformed RTNet in all categories except Grade 3, confirming the pixel-guided fusion module’s role in discriminating benign/malignant features and optimizing ambiguous malignancy classification.

RTNet.S.M and RTNet.C.M: In RTNet.C.M, a decrease in accuracy for grade 3 pulmonary nodules was observed, confirming the unique ability of median pooling to suppress high-frequency CT artifacts, which particularly affect ambiguous malignant lesions. Conversely, the decrease in AUC for the same category in RTNet.S.M reveals how spatial median operations preserve key morphological boundaries of ground-glass opacities, while traditional average/max pooling homogenizes texture gradients. In high-risk stratification, the reduction in W-Acc for RTNet.C.M demonstrates the superiority of median pooling in pulmonary nodule malignancy risk assessment. These findings collectively validate the contribution of median features extracted from feature tensors through median pooling to pulmonary nodule malignancy risk assessment. The TMFM module achieves uncompromised feature fidelity through median pooling, proving that median pooling plays an indispensable role in pulmonary nodule malignancy assessment.

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a).



b).

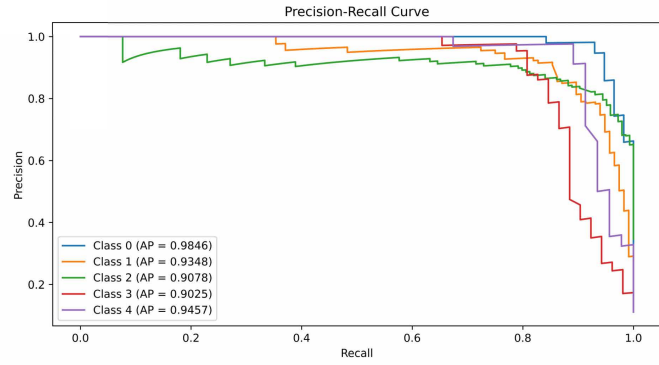


Fig. S-I. a): RTNet's ROC curves and AUC . b): RTNet's Precision-Recall (PR) curves and Average Precision (AP) . Note: the class labels (Class 0–4) shown in the figure correspond to risk levels (Grades

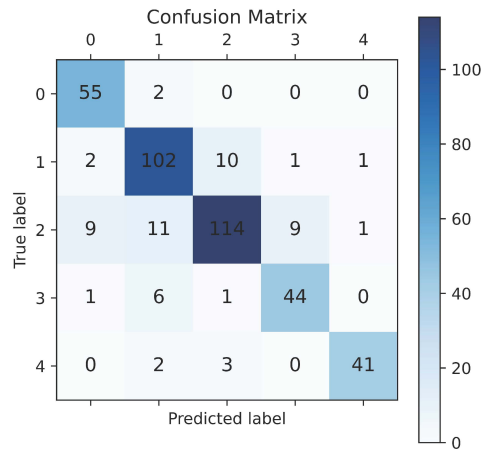


Fig. S-II. Confusion matrix of RTNet on the test set. The matrix presents the classification results across all five risk levels (Grades 1–5). The predicted labels (0–4) and true labels (0–4) in the matrix correspond to risk levels (Grades 1–5), respectively. Diagonal entries represent correctly classified samples, while off-diagonal entries indicate misclassifications between different risk levels.

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